CSWeb 8.1 on cPanel Shared Hosting

Installation and Operation Guide

- About this package
  - Tested configuration
- Before you start
  - Hosting checklist
  - Ask your provider one question
  - What you need
- Installation
  - Step 1. Create the subdomain
  - Step 2. Select the PHP version and settings
  - Step 3. Create the database
  - Step 4. Upload and extract the package
  - Step 5. Enable HTTPS
  - Step 6. Run the setup wizard
  - Step 7. Activate the administration pages
  - Step 8. Validate the installation
- Operation
  - Backups
  - Accounts and security
  - Capacity
  - Where to find logs
- Troubleshooting
- Updating CSWeb
- Support

About this package

This package installs **CSWeb 8.1** (CSWeb Community edition) on a shared hosting account managed with **cPanel**. It is ready to upload: no SSH, Composer or command-line tools are required to install it.

On shared hosting, CSWeb is used for **data collection and synchronization**:

- deploying CSPro applications to tablets and phones,
- synchronizing cases between CSEntry and the server,
- monitoring progress (sync report, map),
- managing interviewers and supervisors,
- downloading the data for processing with CSPro (Data Viewer, export to Stata, SPSS, R or CSV).

The **breakout** feature (automatic export of cases to relational tables) and scheduled jobs are not available on shared hosting. They require a dedicated server.

Tested configuration

The package was installed and tested in an environment reproducing a cPanel shared hosting plan:

Component	Version / setting
PHP	8.3, with `exec`, `proc_open`, `shell_exec`, `system`, `popen` disabled
Web server	Apache 2.4 with `mod_rewrite` and `.htaccess`
Database	**MariaDB 10.6** (also compatible with MySQL 8)
Limits	1 CPU core, 20 database connections per user

Results: installation, application deployment, CSEntry synchronization (including **5,000 cases in a single sync**), data download, user management and Arabic text all work. **200 devices synchronizing within the same minute completed with no error.**

Before you start

Hosting checklist

Requirement	Where to check in cPanel
PHP **8.3** (8.1 minimum)	*MultiPHP Manager*
PHP extensions: `pdo_mysql`, `zip`, `curl`, `openssl`, `fileinfo`, `dom`	*Select PHP Version* or provider
MariaDB 10.6+ or MySQL 8.0+	*MySQL Databases*
Apache `mod_rewrite` and `.htaccess`	Provider (enabled on standard cPanel)
SSL certificate for the subdomain	*SSL/TLS Status* (AutoSSL)
Disk space: about 200 MB, plus data	*Disk Usage*
Files (inodes): about 16,000	Provider
Terminal or SSH access (one command, Step 7)	*Terminal*. Optional, see Step 7

Ask your provider one question

> *What is the maximum number of simultaneous connections per database user (`max_user_connections`) on our plan?*

A value of **20 or more** is suitable. Each synchronization uses one connection for a fraction of a second.

What you need

- The package `csweb-<version>-cpanel.zip` and its checksum file `.sha256`.
- A subdomain name, for example `csweb.your-domain.org`. This guide uses this example.
- A strong administrator password (12 characters or more).

Installation

Step 1. Create the subdomain

1. In cPanel, open **Domains** (or **Subdomains**) and create `csweb.your-domain.org`.
2. Set its **document root** to `/home/<cpanel-user>/csweb`.

   Do not use a folder inside `public_html`. The package is extracted into `/home/<cpanel-user>/csweb` in Step 4.

Step 2. Select the PHP version and settings

1. Open **MultiPHP Manager**, select the subdomain and choose **PHP 8.3**.
2. Open **MultiPHP INI Editor**, select the subdomain and set:

Setting	Value
`memory_limit`	`256M`
`upload_max_filesize`	`64M`
`post_max_size`	`64M`
`max_execution_time`	`300`

Step 3. Create the database

1. Open **MySQL Databases**.
2. Create a database, for example `csweb`. cPanel adds your account prefix: note the full name, for example `myaccount_csweb`.
3. Create a user, for example `myaccount_cswebuser`, with a strong password.
4. Under **Add User To Database**, select the user and the database, then grant **ALL PRIVILEGES**.

Step 4. Upload and extract the package

1. Open **File Manager** and go to your **home folder** (`/home/<cpanel-user>`), not `public_html`.
2. Click **Upload** and select `csweb-<version>-cpanel.zip`.
3. Back in File Manager, right-click the zip and choose **Extract**. This creates the folder `csweb`.
4. Click **Settings** (top right) and tick **Show Hidden Files (dotfiles)**. Open `csweb` and check that the file `.htaccess` is present. Without it, CSEntry cannot authenticate.

5. Delete the zip file.

> **Optional: check the download.** Compare the SHA-256 of the zip with the `.sha256` file before uploading (on Windows: `certutil -hashfile file.zip SHA256`).

Folder permissions are already correct: on cPanel, PHP runs as your account, so **do not set 777 permissions**.

Step 5. Enable HTTPS

1. Open **SSL/TLS Status**, select `csweb.your-domain.org` and click **Run AutoSSL**.
2. Wait until `https://csweb.your-domain.org` opens without a certificate warning. A page from the CSWeb setup is expected at this stage.

Complete this step **before** Step 6: the address entered in the setup wizard is the one written into the CSEntry applications.

Step 6. Run the setup wizard

1. Open `https://csweb.your-domain.org/setup/`. All requirements must show a green check. If one is red, see *Troubleshooting*.
2. Click **Next** and fill in the form:

Field	Value
Database name	Full name from Step 3, for example `myaccount_csweb`
Database host	`localhost`
Database user / password	From Step 3
Admin password	Your strong administrator password
Timezone	Your local timezone, for example `Asia/Baghdad`
Max execution time	`300`
Files directory	`/home/<cpanel-user>/csweb/files` (pre-filled)
CSWeb API URL	`https://csweb.your-domain.org/api/` **(with the final slash)**

3. Click **Next**. The wizard creates the database and shows a completion page.

After installation, the wizard **locks itself** (`/setup/` returns *403 Forbidden*). This is expected and protects your installation.

Step 7. Activate the administration pages

This step enables the **Settings**, **Backup** and **Logs** pages. Data collection and synchronization work without it.

1. In cPanel, open **Terminal**.
2. Run this command, replacing `ea-php83` with the PHP version chosen in Step 2 if it differs:

```
cd ~/csweb && /opt/cpanel/ea-php83/root/usr/bin/php bin/console csweb:community:install-schema --env=prod
```

3. The expected result ends with: `[OK] Community schema installed`.

If **Terminal** is not available on your plan, ask your provider to run the command once, or skip this step.

Step 8. Validate the installation

Complete this checklist before using the server for fieldwork.

#	Test	Expected result
1	Log in at `https://csweb.your-domain.org` as `admin`	The dashboard opens, with styled tables and menus
2	**Users**: create an interviewer account	The account appears in the list
3	From **CSPro Designer**, deploy an application to `https://csweb.your-domain.org/api`	The application appears under **Apps**
4	On a device, in **CSEntry**, add the application from the server, log in as the interviewer	The application installs
5	Enter 2 or 3 test cases and **synchronize**, preferably over mobile data	Sync completes; cases appear in **Sync Report**
6	Synchronize a second device	It downloads the shared files and data as designed
7	Download the data with **CSPro Data Viewer** (download from CSWeb, using the server address above)	All test cases are present

Then delete the test data from the dashboard's list of dictionaries (*Delete data*), so that fieldwork starts from an empty database.

Operation

Backups

Use the **cPanel backups**. CSWeb's own *Backup now* button needs functions that shared hosting disables, and shows an error.

- **Backup** (or **JetBackup**) in cPanel saves the database and files daily on most plans. Check the frequency and retention with your provider.
- During fieldwork, download a copy at least weekly: **Backup > Download a MySQL Database Backup** (your CSWeb database) and **Download a Home Directory Backup**.
- Data is also kept on the devices until it is synchronized. Do not delete CSEntry data on devices until it has been downloaded and checked.

Accounts and security

- Use one account per interviewer. The server limits repeated wrong passwords: after **5 failed attempts in one minute**, an account is locked for one minute.
- Keep the `admin` password private. Create supervisor accounts for daily monitoring.
- Always use the `https://` address in CSPro Designer and CSEntry.

Capacity

Situation	Behaviour
Up to 200 devices synchronizing within the same minute	Tested with no error
More than 400 devices within the same minute	A few syncs may fail and must be retried
Devices sharing one network (office Wi-Fi, mobile operator NAT)	Supported (each interviewer has their own account)
Large syncs (5,000 cases at once)	Tested: about 2 seconds

For a census or a survey with several hundred interviewers synchronizing at the same time, a dedicated server is recommended.

Where to find logs

`/home/<cpanel-user>/csweb/var/logs/` (`api-<date>.log` for synchronization, `ui.log` for the web interface). These files are not accessible from the web.

Troubleshooting

Symptom	Cause and solution
A requirement is red in the setup wizard	PHP version or extension: see Step 2 and the hosting checklist. For “directories are writeable”, re-extract the zip with the File Manager.
HTTP 500 when logging in to the dashboard right after setup	The API URL lacks its final slash. In File Manager, edit `csweb/src/config.php` and make the `API_URL` line end with `/api/'`.
Pages display without style; buttons and tables do not work	Incomplete upload. Delete the `csweb` folder and extract the full zip again.
CSEntry or Designer: *Unauthorized* (401) with correct passwords	The hidden file `csweb/.htaccess` is missing. Re-extract the zip (Step 4).
Too many requests (429)	Repeated wrong passwords: wait one minute.
Some syncs fail at peak times; the log shows `max_user_connections`	The plan’s connection limit is reached. Ask the provider to raise it, or schedule teams to sync at different times.
Large syncs fail or time out	Check Step 2 settings (`max_execution_time`, upload limits). If the provider’s firewall (ModSecurity) blocks requests, ask them to allow `/api/` on the CSWeb subdomain.
Backup now shows an error	Expected on shared hosting: use the cPanel backups.
Settings, Backup or Logs page shows an error	Step 7 was not run.
`/setup/` returns *403 Forbidden*	Normal after installation.

Updating CSWeb

1. Make a cPanel backup (database and home directory).
2. Extract the new package into a new folder, for example `csweb-new`.
3. Copy into it, from the current installation: the file `src/config.php` (written by the setup wizard) and the folders `files/` and `files_csweb/`.
4. Swap the folders (rename `csweb` to `csweb-old`, then `csweb-new` to `csweb`), and empty the folder `csweb/var/cache`.
5. Repeat Step 7 and the checklist of Step 8. Keep `csweb-old` until the update is validated.

Support

CSWeb Community: https://github.com/BOUNADRAME/csweb-community. Report issues under *Issues* with the content of `BUILD-INFO.txt` and the relevant lines of `var/logs/`.

Installation assistance and dedicated server set-up are available from StatPro Consulting.